

- FC neural networks MLP

- Convolutional Neural Networks, 2012-2014
"superhuman" for image classification

- Sequential data, Natural language text
NLP - processing of $\vec{v}$

Ex.: Time series forecasting



$\hat{y}(\vec{x}_t) \approx y_{t+1}$
Linear regression / Gradient Boosting

$\vec{x}_t = ?$

$$\vec{x}_t = (y_t, y_{t-1}, \dots, y_{t-k})$$

$$\hat{y}_{t+1} = \sum_{i=0}^k W_i y_{t-i}$$

[A No cat jumps over a lazy dog]
input $\vec{x}$ $\vec{y}$
200 000 - dictionary

$$O(d \times d)$$

$$O(d \times k)$$

one-hot encoder $\begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} \leftarrow \text{index of "lazy"}$

Predict the next word (token) PNW

$$\vec{x} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 2 \\ 1 \\ 0 \\ 1 \\ 1 \\ 0 \\ 1 \end{pmatrix} \begin{matrix} \text{"a"} \\ \vdots \\ \text{cat} \\ \text{"lazy"} \end{matrix}$$

$$b(W\vec{x}) = \hat{\vec{p}}$$

1. We don't take order into account
2. W is large d^2

$$\text{PNW} = \text{LM}$$

$$\frac{P(y_{t+1} | y_1, \dots, y_t)}{P(y_1, \dots, y_t, y_{t+1})} - ?$$

FC NN - doesn't work

CNN
word2vec/
Gemma

$$y_1 \dots y_t \rightarrow \vec{x}_1^{(1)} \dots \vec{x}_k^{(1)}$$

$$y_t \rightarrow \vec{x}_t \in \mathbb{R}^d$$

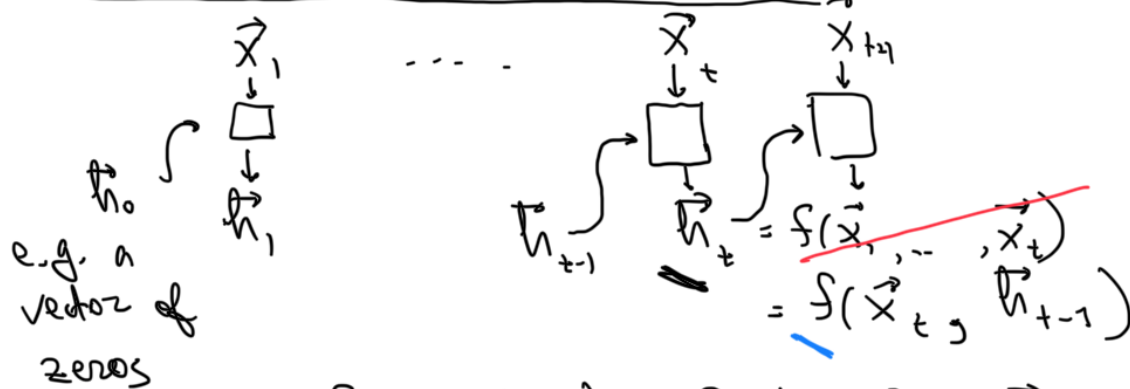


1-D conv. layer

$$\vec{x}_i^{(e+1)} = \sum_{j=-k}^{k} w_j \vec{x}_{i+j}^{(e)}$$

$(2k+1)$ parameters

Recurrent neural networks



various RNN

LSTM

long-short term memory

GRU

gated recurrent

$$f(\vec{x}_t, \vec{h}_{t-1}) = \sigma(W_x \vec{x}_t) \odot \vec{x}_t + (1 - \sigma(W_x \vec{x}_t)) \odot \vec{h}_{t-1}$$

$$f(\vec{x}_t, \vec{h}_{t-1}) = \sigma(W_x \vec{x}_t) \odot \sigma(W'_x \vec{x}_t) + \text{unit} + (1 - \sigma(W_x \vec{x}_t)) \odot \vec{h}_{t-1}$$

Backpropagation: we multiply t terms
 $\text{terms} > 1 \Rightarrow$ gradient explosion
 $< 1 \Rightarrow$ gradient vanishes

1. No long term memory - gradients fade

2. Bottleneck: all info is in $\vec{h}_t \in \mathbb{R}^d$

